



TalkReal

Sound natural. Speak with confidence.

CS 185 Final Report | Winter 2026 | Victor P. and Ryan C.

Problem

Speaking is the strongest signal of whether someone feels they know a language, but it is also the hardest skill to practice consistently and confidently.

Solution

TalkReal gives learners a low-pressure place to practice speaking with AI, receive naturalness-focused feedback, and build confidence for real conversations.

Needfinding • Experience Prototypes • Design Evolution • Final Prototype • Reflection

1. Project Name and Value Proposition

TalkReal is a mobile speaking practice app for language learners who want to sound conversationally natural in real-world situations. Instead of focusing mainly on grammar drills or pronunciation scores, TalkReal helps users practice speaking out loud, receive naturalness-focused feedback, and leave each session with clearer phrasing, more confidence, and a stronger sense of what people would actually say in context.

2. Team Members and Roles

- Victor P. - research, interaction design, prototyping, writing
- Ryan C. - research, visual design, prototyping, usability testing

3. Problem and Solution Overview

Speaking is often the skill people use to judge whether someone "really knows" a language, yet it is also one of the hardest skills to practice consistently. Learners need a low-pressure environment where they can speak freely, make mistakes safely, and still gain feedback they trust for real conversations.

TalkReal addresses this gap through short spoken activities, free-response practice, and style-based conversation practice with AI. After each activity, the app provides feedback on conversational naturalness, suggests more natural alternatives, and encourages a repeatable loop of speak, reflect, and try again.

4. Needfinding

4a. Interviews

Our needfinding process took place across GR1 and GR2 and included five core interviewees representing a mix of extreme users, average users, and non-users. We intentionally sought variety in language background, comfort with AI, and the social contexts in which speaking mattered most. Participants included learners who relied on English for day-to-day independence, learners using AI regularly for Japanese or Spanish practice, and non-users who still struggled with the emotional difficulty of speaking in public.

We recruited participants through a combination of informal intercepts and personal outreach in places such as cafes, parks, and everyday public settings, plus a small number of remote conversations when that was more practical. In the earlier interviews, our team used a simple two-person structure with one interviewer and one scribe so we could capture direct quotes, emotional cues, and follow-up questions in real time. Based on our documented materials, these early needfinding interviews were informal course-project conversations and were not run as compensated research sessions.

Our questions focused less on language correctness in the abstract and more on lived experience: when learners feel nervous speaking, what they trust or distrust about AI, when they avoid speaking entirely, and what kinds of practice actually feel useful. This helped us move away from a broad "AI for language learning" space and toward a tighter focus on conversational speaking, naturalness, confidence, and real-world transfer.

4b. Synthesis

After interviewing, we synthesized our findings through participant-specific empathy work and cross-participant theme finding. Rather than flattening everyone into one generic user, we compared how different learners experienced judgment, hesitation, trust, and motivation. That process led us to a few durable themes:

- Low-pressure practice increases willingness to speak.
- Learners value AI because it is always available and non-judgmental.
- Naturalness matters more than textbook correctness when the goal is real conversation.
- Trust breaks down when feedback feels too formal, too generic, or disconnected from real life.
- Pace matters: users want time to think, pause, and revise without social pressure.
- Real scenarios such as small talk, appointments, ordering, and directions feel more motivating than generic chat.

These findings gave us a sharper design direction. The need was not simply "practice speaking with AI." The need was to help learners practice spoken language in a way that protects confidence while still validating whether their speech sounds natural enough to use in the real world.

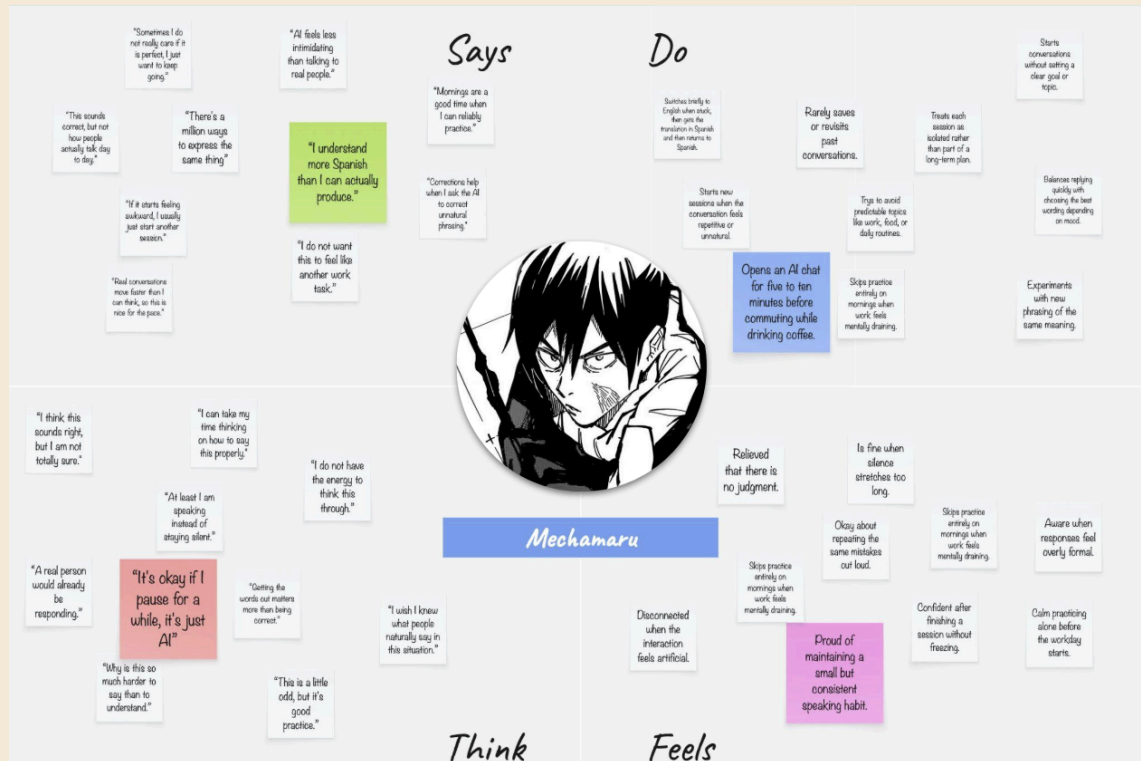


Figure 1. Example synthesis artifacts from needfinding and empathy mapping. The first board captures Mechamaru's speaking habits, hesitations, and motivations.

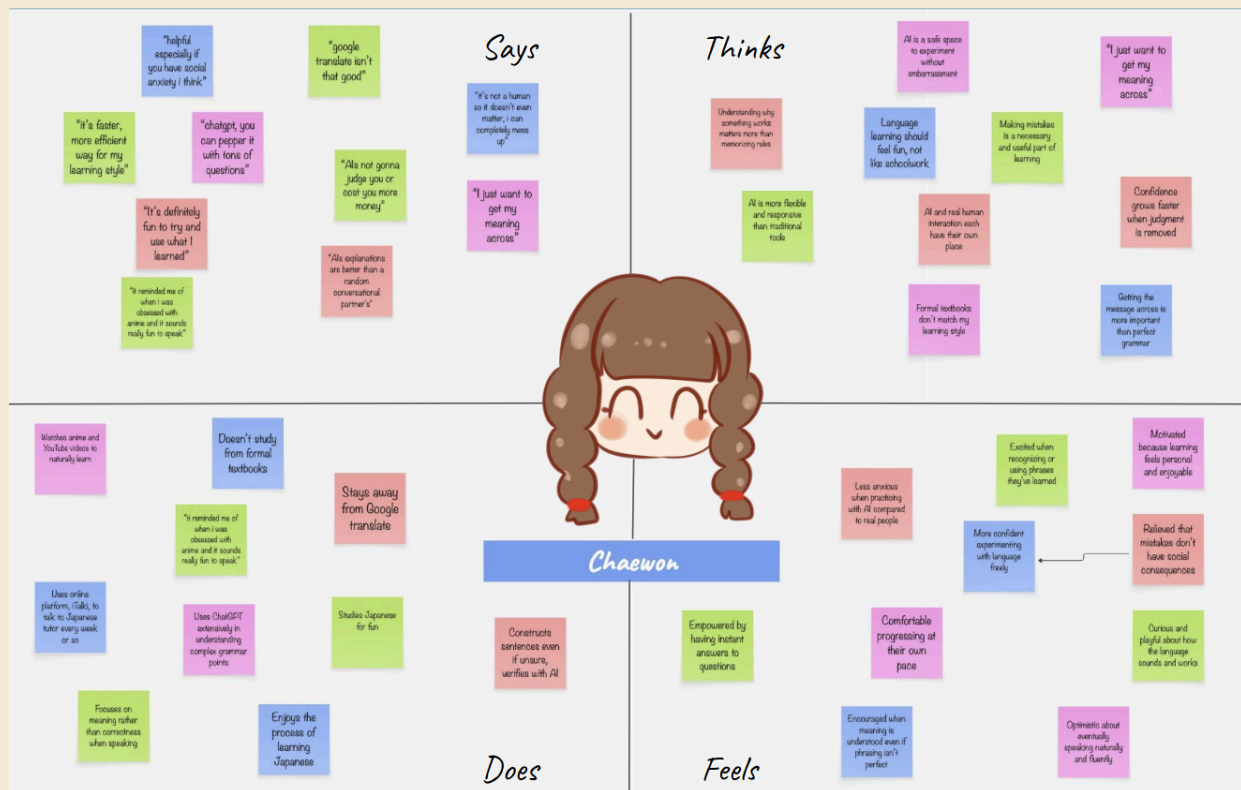


Figure 2. A second empathy-style synthesis board shows how Chaewon's thoughts, behaviors, and emotions reinforced themes of confidence, experimentation, and trust in AI.

5. POVs and Experience Prototypes

5a. Final POV Statements

leiri POV. leiri is a 63-year-old retired art teacher who learned English largely through daily life and still feels anxious in public speaking situations. She needs a safe way to rehearse high-stakes everyday conversations because emotional risk, not vocabulary alone, is what prevents her from speaking confidently.

Mechamaru POV. Mechamaru is a 24-year-old software engineer who occasionally practices spoken Spanish with AI before work. She needs spoken practice that lets her pause, think, and revise at her own pace because lower pressure helps her experiment and speak more consistently.

Amanai POV. Amanai is a student who already uses English daily and turns to AI because it feels safe and always available. She needs reassurance that what she practices would sound natural in real life because confidence depends on whether the language will actually work outside the app.

5b. Sampling of HMWs

From leiri:

- How might we reduce fear of making mistakes when talking to strangers?
- How might we create practice that feels realistic but still non-judgmental?
- How might we support a gradual path from typing practice to speaking practice?

From Mechamaru:

- How might we let learners pause and think while speaking?
- How might we support stress-free spoken practice?
- How might we make AI speech sound casual and natural?

From Amanai:

- How might we help learners know if people would actually say this?
- How might we validate spoken practice without adding pressure?
- How might we bridge safe AI practice and real-life usage?



Figure 3. A broad HMW board helped us expand the design space around confidence, realism, and lower-pressure speaking.

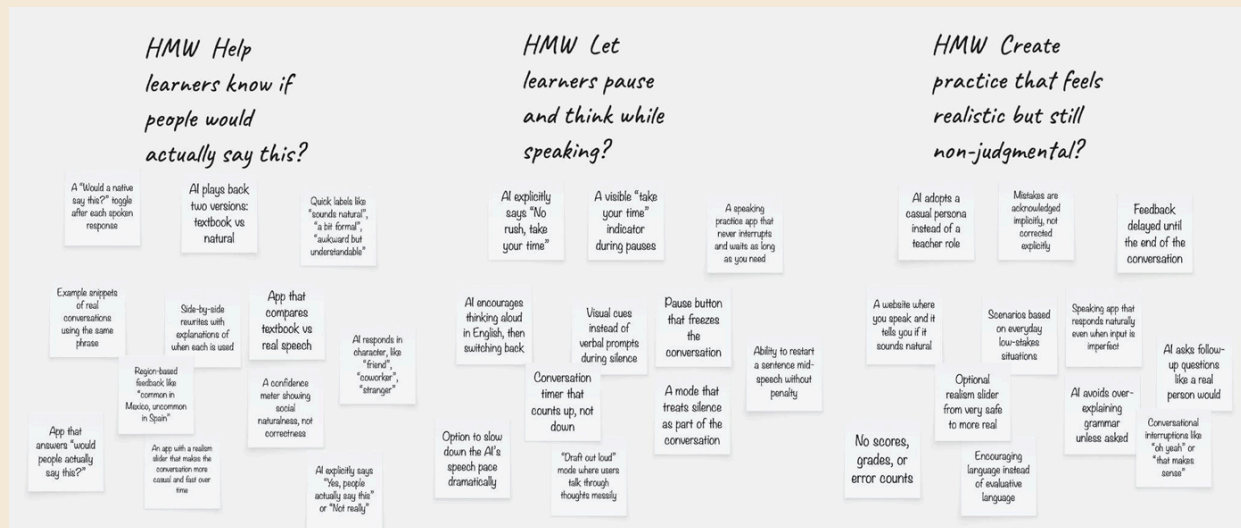


Figure 4. A more focused brainstorming board translated three high-priority HMWs into many candidate product directions and interaction ideas.

5c. Top Three Solutions from Brainstorming

1. **A realism-slider conversation app.** Practice starts highly supportive, then gradually becomes faster and more realistic.
2. **An unlimited-wait speaking practice app.** The partner never interrupts, allowing the learner to pause as long as needed.
3. **A naturalness-validation speaking app.** The learner speaks, and the system tells them whether the phrasing sounds conversationally natural, with light feedback and examples.

5d. Experience Prototypes

Prototype 1: Realism Slider Conversation App

Assumption being tested: Gradually increasing realism would build confidence better than starting with a fully realistic conversation.

Setup: Two participants role-played the same scenario three times. Round one was slow and highly supportive. Round two was more natural. Round three was faster, more casual, and introduced mild real-world friction.

What worked: Participants liked starting in a safe mode and reported more confidence by the later rounds.

What did not work: The jump to the hardest round still felt too sudden for some people.

Implication: Realism should not be binary. Confidence-building works best when difficulty ramps up in smaller, learner-controlled steps.

Prototype 2: Unlimited Wait Speaking Practice

Assumption being tested: If the system never interrupts and allows long pauses, learners will speak more and feel less anxious.

Setup: Participants completed the same speaking prompt twice. In one version, the partner waited through silence and never interrupted. In the other, the partner responded with more typical conversational timing.

What worked: Participants spoke longer and felt more in control when interruptions were removed.

What did not work: Complete silence occasionally felt awkward without reassurance that pausing was acceptable.

Implication: Removing time pressure is powerful, but the interface still needs to communicate that the learner is allowed to pause and think.

Prototype 3: Naturalness Validation App

Assumption being tested: Simple, non-judgmental validation about naturalness would make users more confident using phrases in later conversation.

Setup: In a face-to-face role-play, one person spoke a phrase and the other responded with lightweight naturalness validation rather than heavy correction.

What worked: Participants reused validated phrases later in the conversation without hesitation.

What did not work: Binary validation was sometimes too thin; a few users wanted a brief reason or context.

Implication: This concept struck the best balance between confidence, clarity, and usability. It became the foundation for TalkReal because it addressed a validated need without adding pressure.

6. Design Evolution

6a. Final Solution

Our final solution is a mobile app that lets learners practice spoken language through three task types and then receive feedback centered on conversational naturalness. The system does not just say whether a phrase is technically correct. It shows users where they sounded stiff, overly formal, unclear, or mismatched to context, then suggests better alternatives and next steps.

We chose this direction because it consistently aligned with our evidence across interviews and experience prototypes. Learners told us they avoided speaking when they feared judgment, wanted more confidence before real interactions, and cared deeply about whether a phrase sounded like something a real person would say. Our experience prototypes then showed that lightweight validation, flexible pacing, and a sense of safety meaningfully changed how willing participants were to keep speaking. That combination made TalkReal more compelling than a general conversation bot or a standard language-learning drill app.

Before we locked into the final mobile direction, we also explored alternative form factors and surrounding product directions. The strongest early concepts centered on phone, AR, and AirPods-based experiences, but we ultimately chose phone because it offered the lowest barrier to adoption and the clearest way to present visible, phrase-level feedback. We also compared TalkReal against adjacent speaking tools to clarify what would differentiate our concept: naturalness feedback, emphasis on real-life use cases, and spoken interaction designed for confidence rather than just correctness.

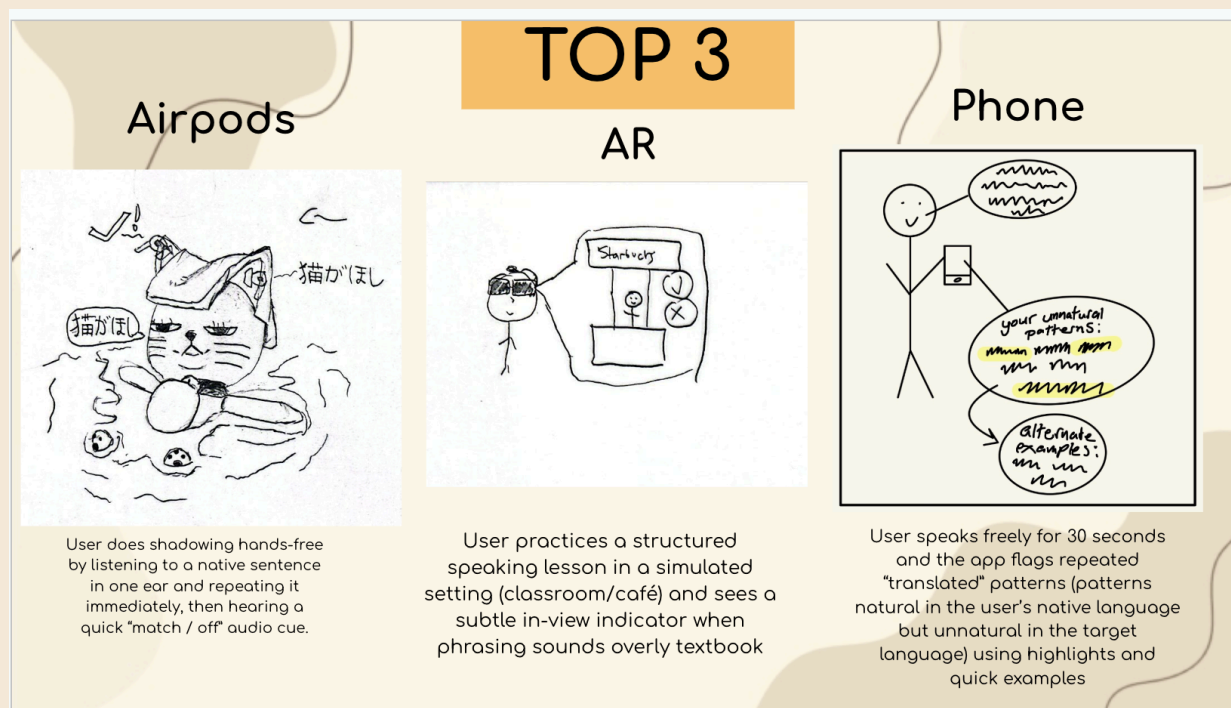


Figure 5. Early concept exploration compared AirPods, AR, and phone-based directions before the team converged on a mobile-first solution.

	talkreal			
instant responses	✓	✓	✓	✓
caters to multiple languages	✓	✓		✓
specifically for speaking naturally	✓			
emphasis on real-life use cases	✓			✓

Figure 6. A lightweight market-research comparison helped position TalkReal against adjacent speaking tools and sharpened the product's differentiation.



Figure 7. One storyboard segment from the concept video shows TalkReal signaling successful phrasing and reinforcing the user's growing speaking confidence.

6b. Tasks

Simple Task: Short Prompt

The user answers a short scenario-based prompt within a one-minute window and receives targeted feedback.

Why it matters: This is the easiest entry point into the product. It gives anxious users a bounded, low-pressure speaking activity that still produces immediate value.

Moderate Task: Free Response

The user speaks freely about any topic, at their own pace, and then receives a more comprehensive naturalness analysis.

Why it matters: This supports users who want more authentic speaking practice and the freedom to express themselves without a scripted scenario.

Complex Task: Conversation Practice

The user chooses a conversational style, such as casual, polite, professional, or academic, and has a multi-turn exchange with TalkReal before receiving style-specific feedback.

Why it matters: This task gets closest to real-world transfer. It is especially relevant for languages and settings where tone, register, and social context strongly affect how natural speech sounds.

6b.i. Annotated Task Flows Using the Final Interface

The figures below show representative screens from the final interface. Together they illustrate the core loop of selecting an activity, speaking, reviewing before analysis, and then acting on feedback.

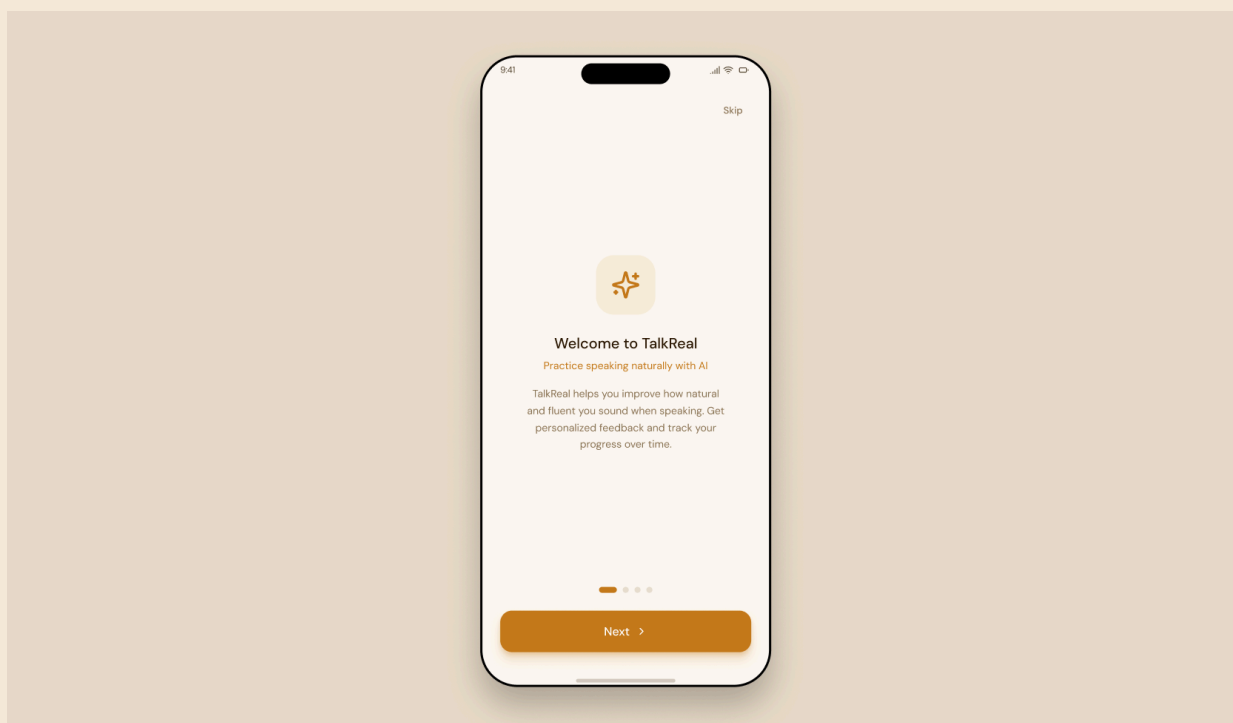


Figure 8. High-fi onboarding introduces TalkReal's purpose and sets expectations for first-time users.

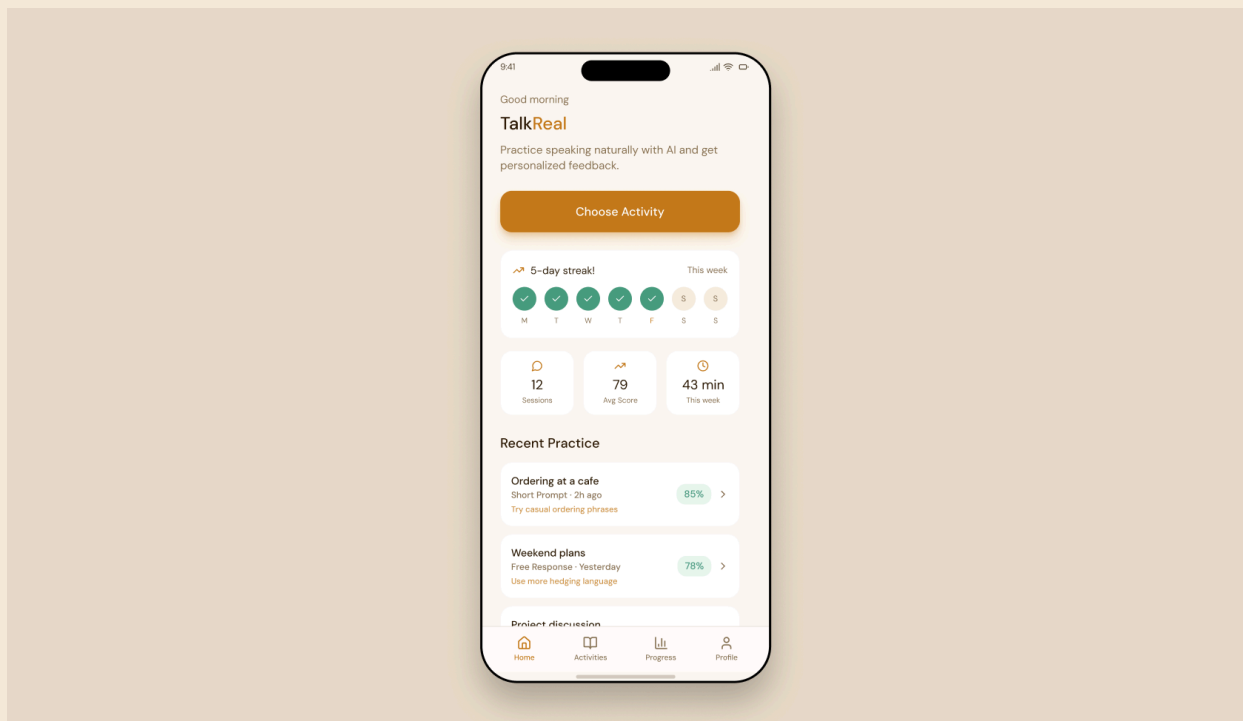


Figure 9. The high-fi home dashboard makes the practice loop visible through recent activity and progress cues.

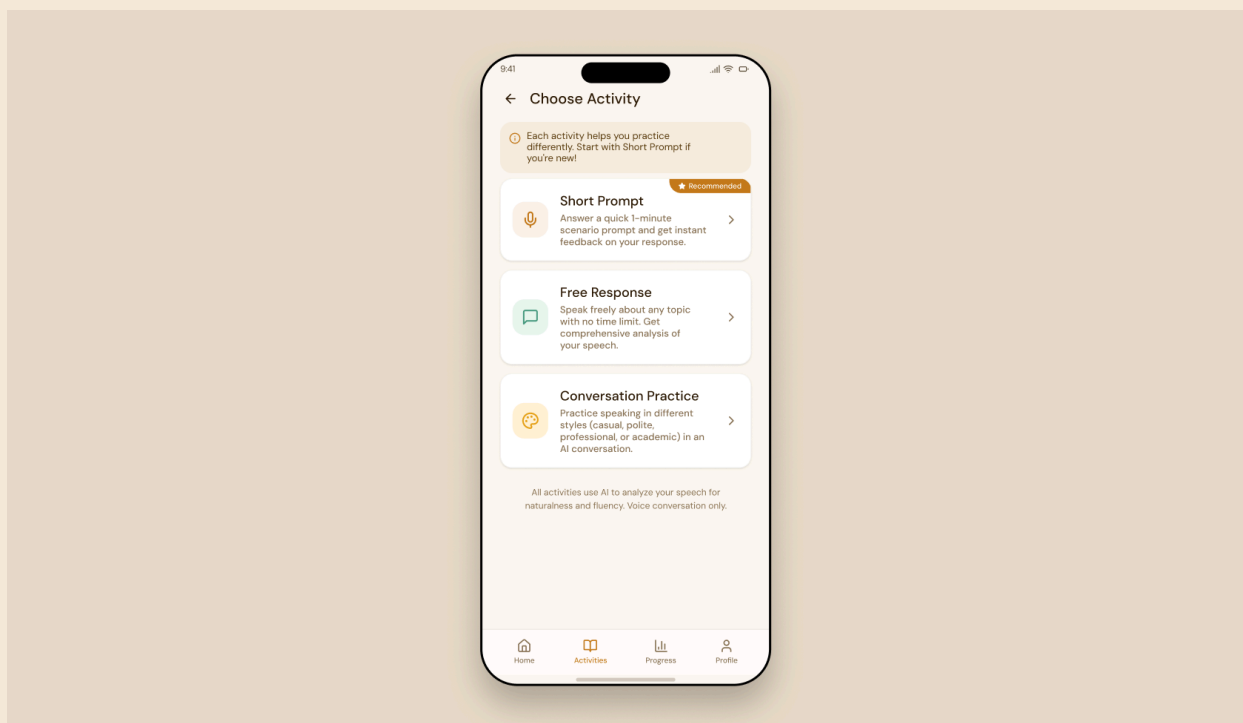


Figure 10. The high-fi activity selection screen clarifies the differences among the three task types.

For the three core tasks, we show the end-to-end flow visually so the sequence is easy to scan at a glance.

Short Prompt Flow

This bounded path gives first-time users a strongly guided entry into the product, from choosing the activity to receiving immediate feedback.

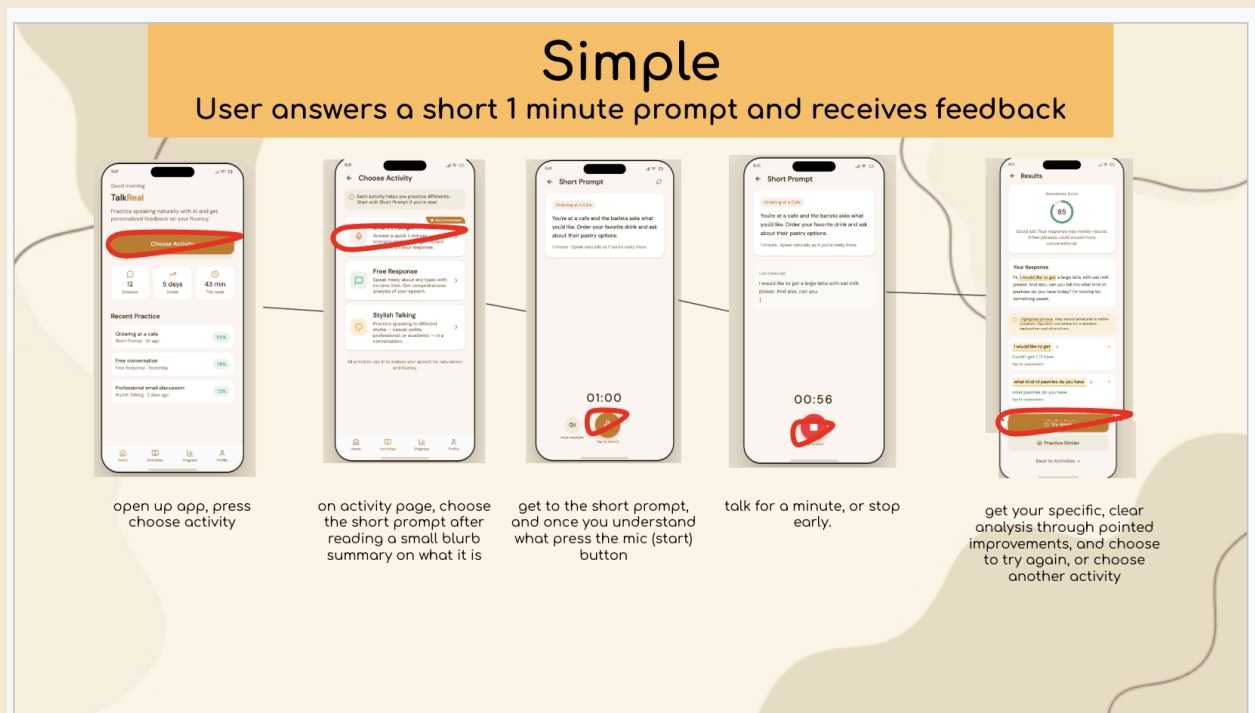


Figure 11. The Short Prompt flow shows the guided path from activity selection to timed recording and clear feedback.

Free Response Flow

This flow keeps the interaction more open-ended while still moving the user through a clear speaking-to-analysis loop.

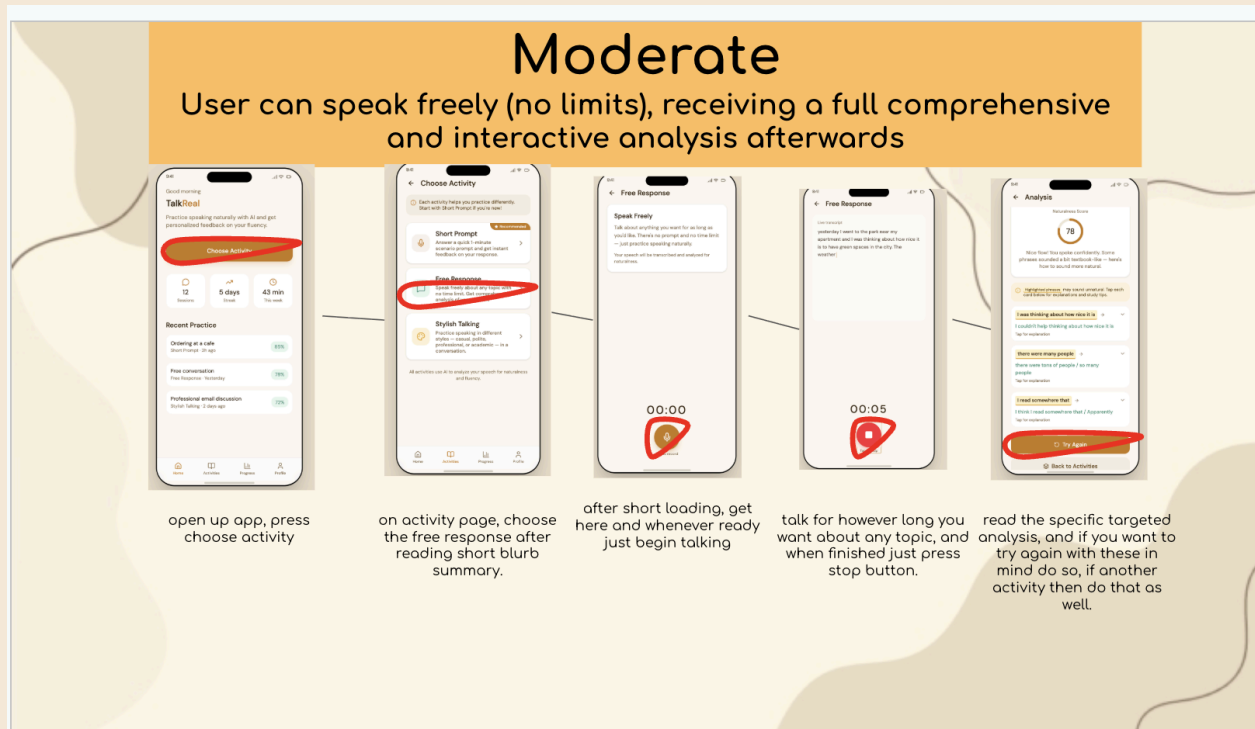


Figure 12. The Free Response flow shows how open speaking leads into fuller post-task analysis and retry.

Conversation Practice Flow

This final flow supports the most complex task, combining style selection, conversation setup, and multi-turn interaction.

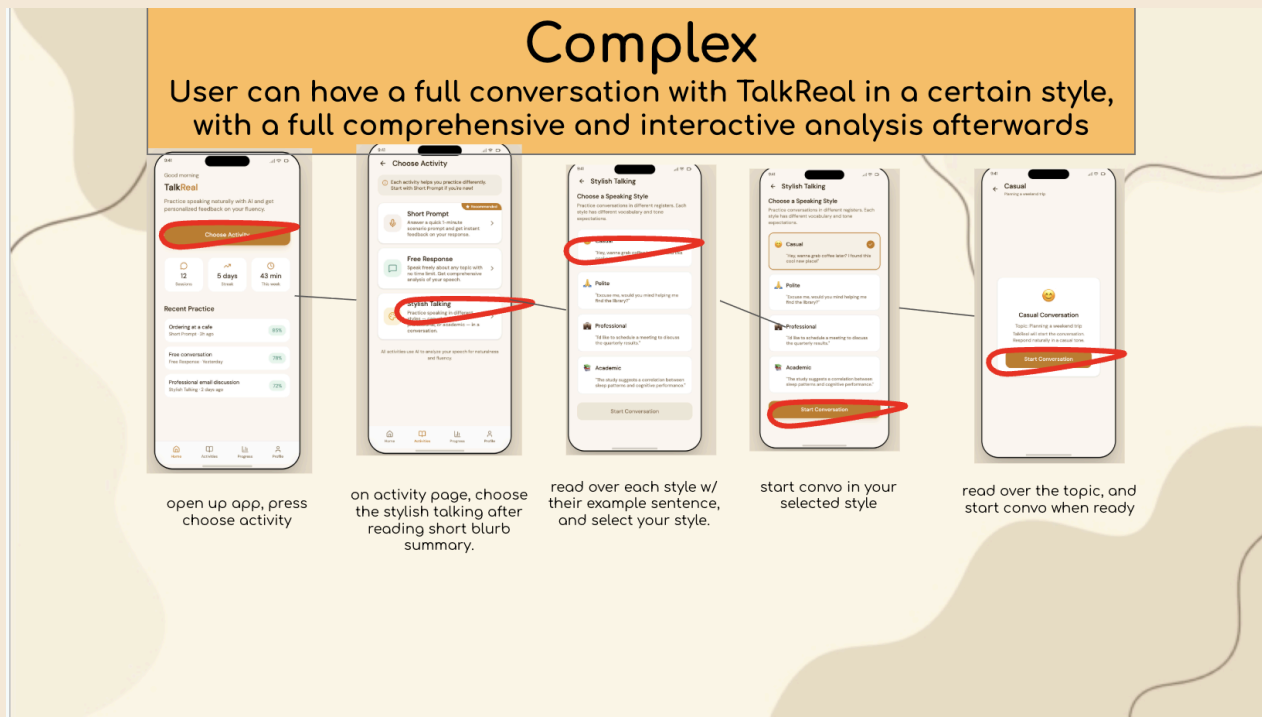


Figure 13. The Conversation Practice flow shows style selection, conversation setup, and the transition into the longer speaking exchange.

6c. Design Evolution Visualizations and Rationale

Low-Fi Evolution and Transition to Medium-Fi

Our low-fi prototype let us test the structure of the experience before investing in visual polish. In GR5, two participants completed all three activities using a paper prototype while one teammate facilitated and the other logged assists, misclicks, and critical incidents. The strongest findings were not about whether users liked the idea. They did. The strongest findings were about whether the flow was self-explanatory on first use.

Three issues stood out in low-fi testing:

4. Users were confused by the difference between Short Prompt, Free Response, and Stylish Talking.
5. Users did not understand that certain interface elements, especially the speaker icon and highlighted phrases, were interactive.
6. Users expected a cleaner and more direct loop from activity selection to speaking to actionable feedback.

These findings led directly to the medium-fi prototype. We added short descriptions under each activity, recommended Short Prompt for first-time users, clarified style options with examples, and made the analysis screen more explicit about what the user could tap and why.

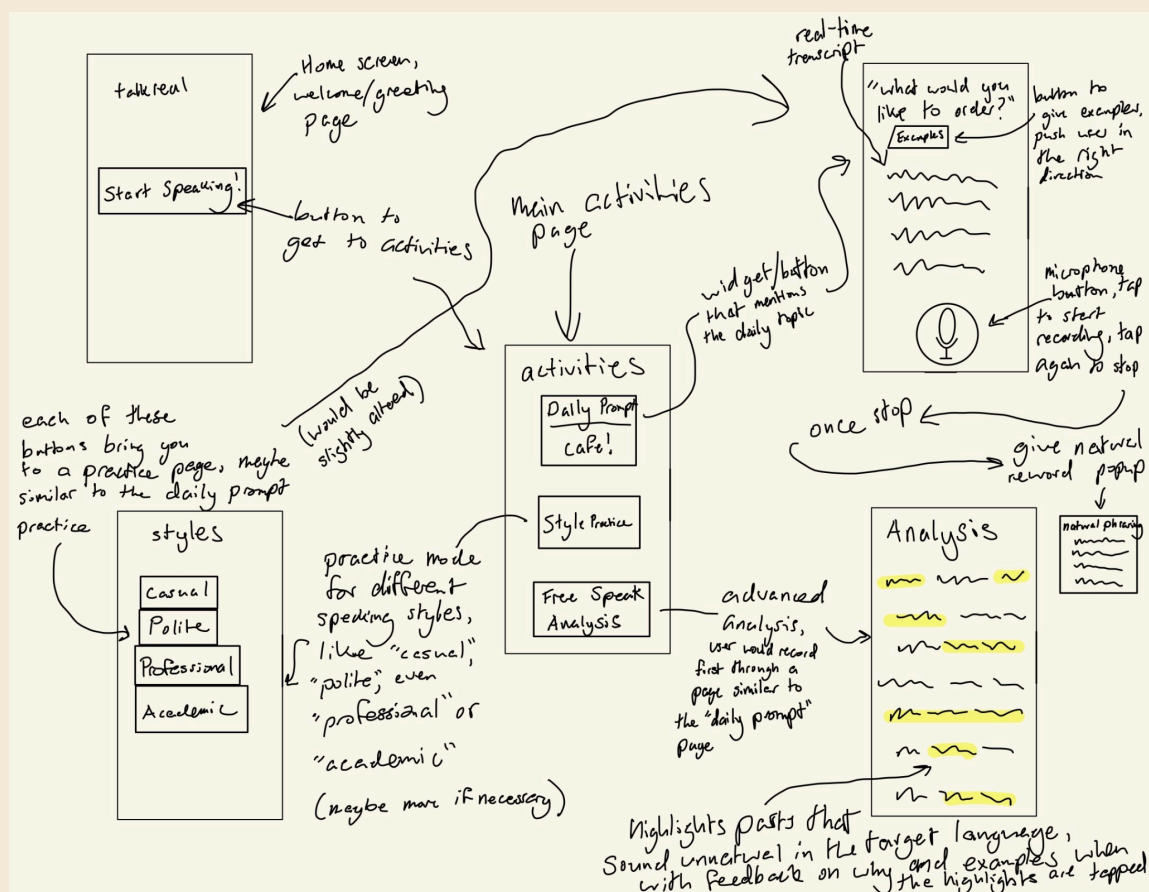


Figure 14. An annotated low-fi flow sketch of the overall TalkReal concept helped the team reason about entry points, activity selection, recording, and analysis during the low-fi phase.

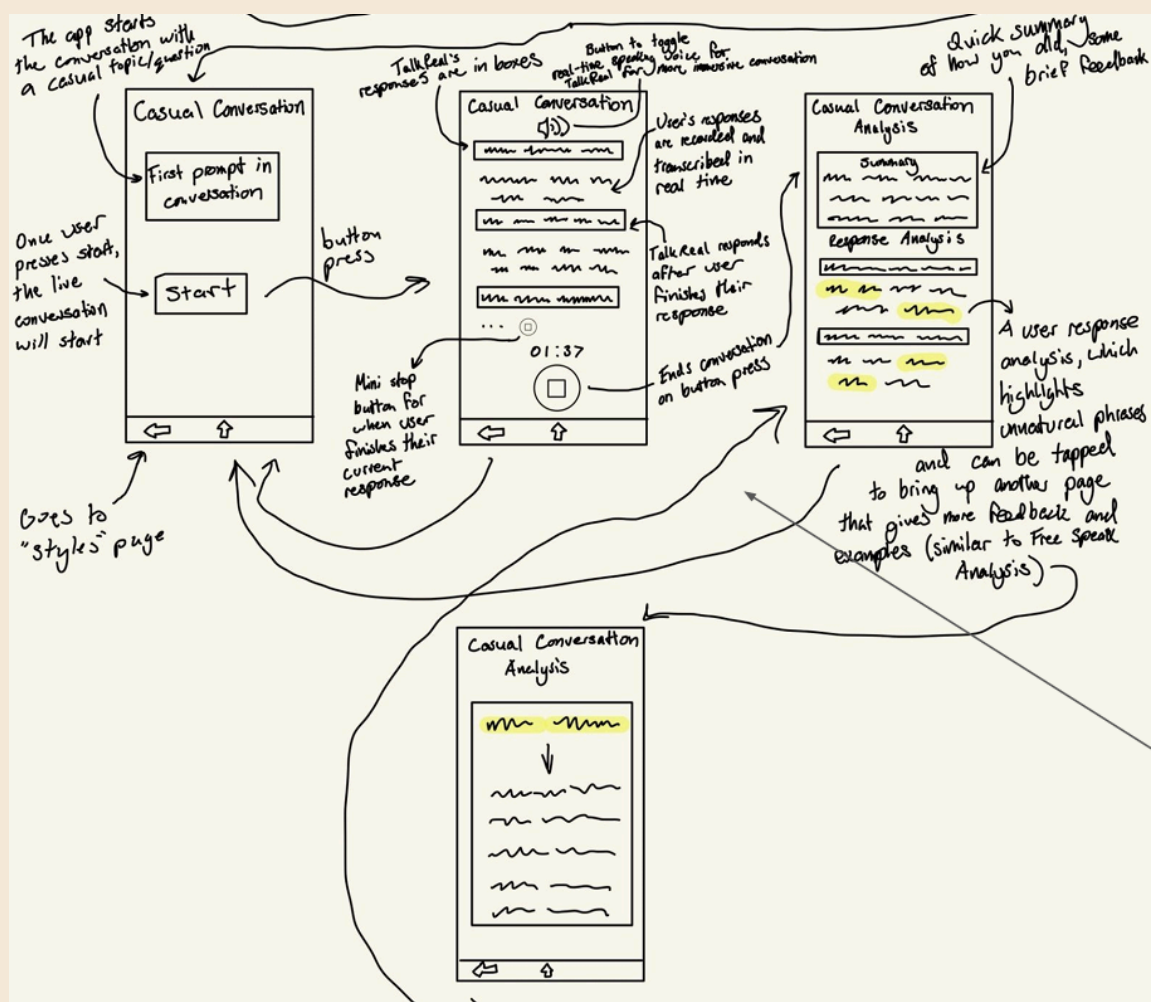


Figure 15. A second annotated low-fi flow focused on conversation-practice structure and the relationship between speaking turns and post-task analysis. This was also part of the low-fi evolution, not the medium-fi prototype.

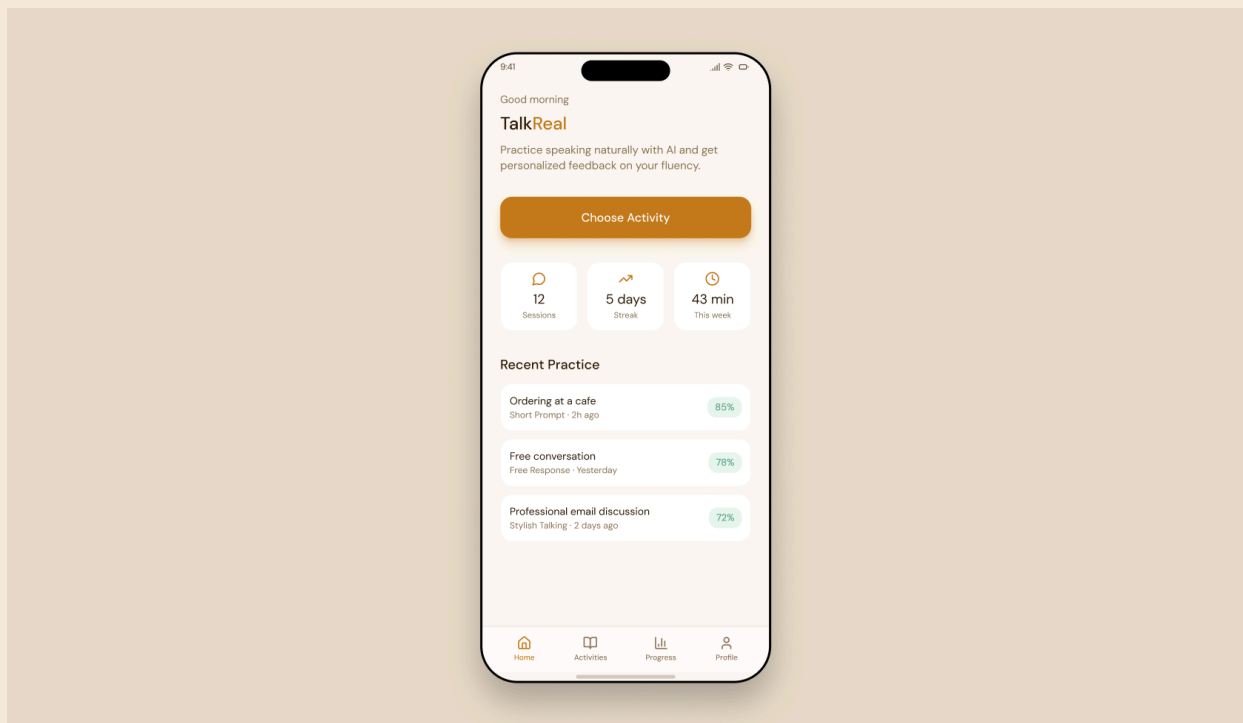


Figure 16. Medium-fi home screen.

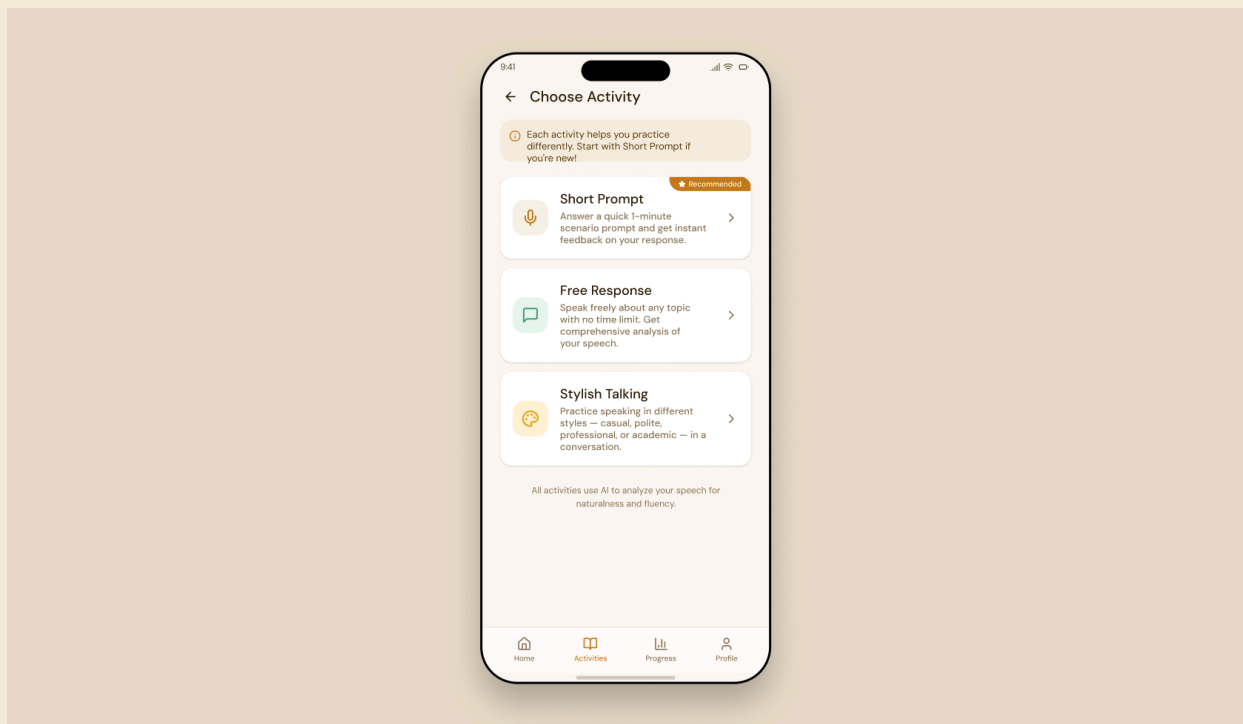


Figure 17. Medium-fi activity selection clarified the task menu after low-fi confusion.

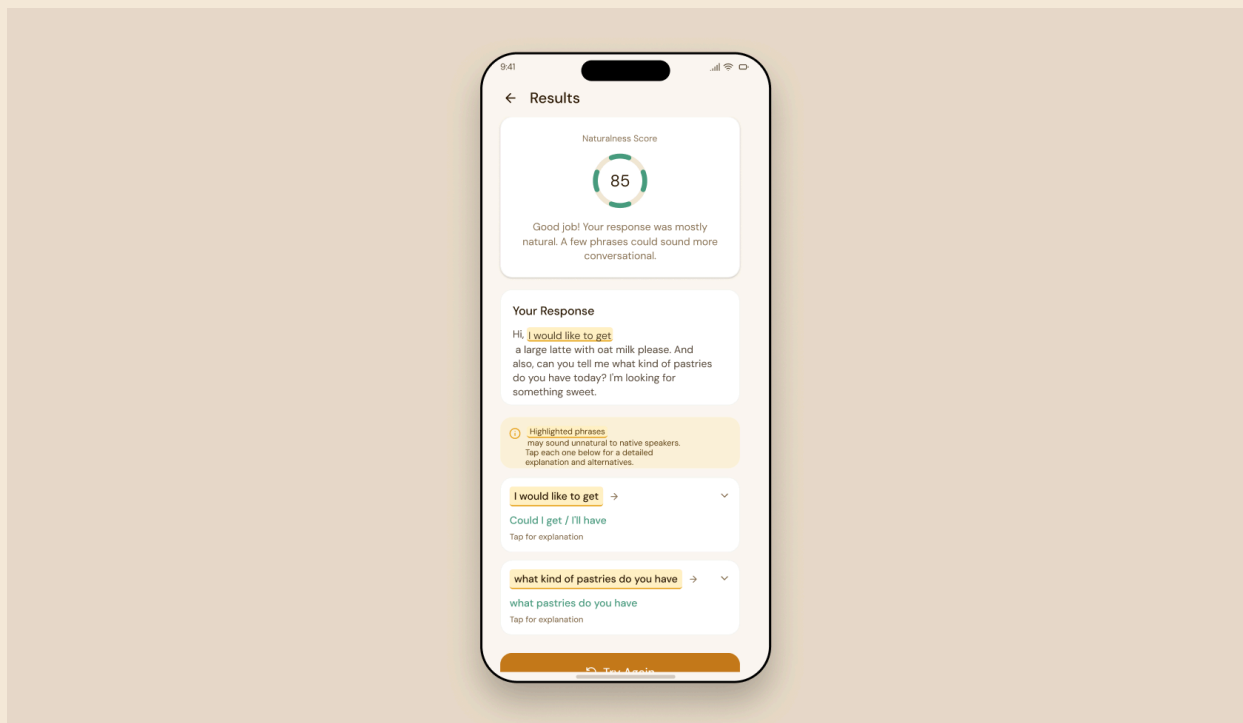


Figure 18. Medium-fi results screen emphasized phrase-level feedback and retry actions.

Methodologically, this stage relied on paper prototyping and Wizard-of-Oz facilitation. What we learned was mainly about information architecture, discoverability, and mental models. The implication was that even a strong concept would fail if the first-use flow remained ambiguous.

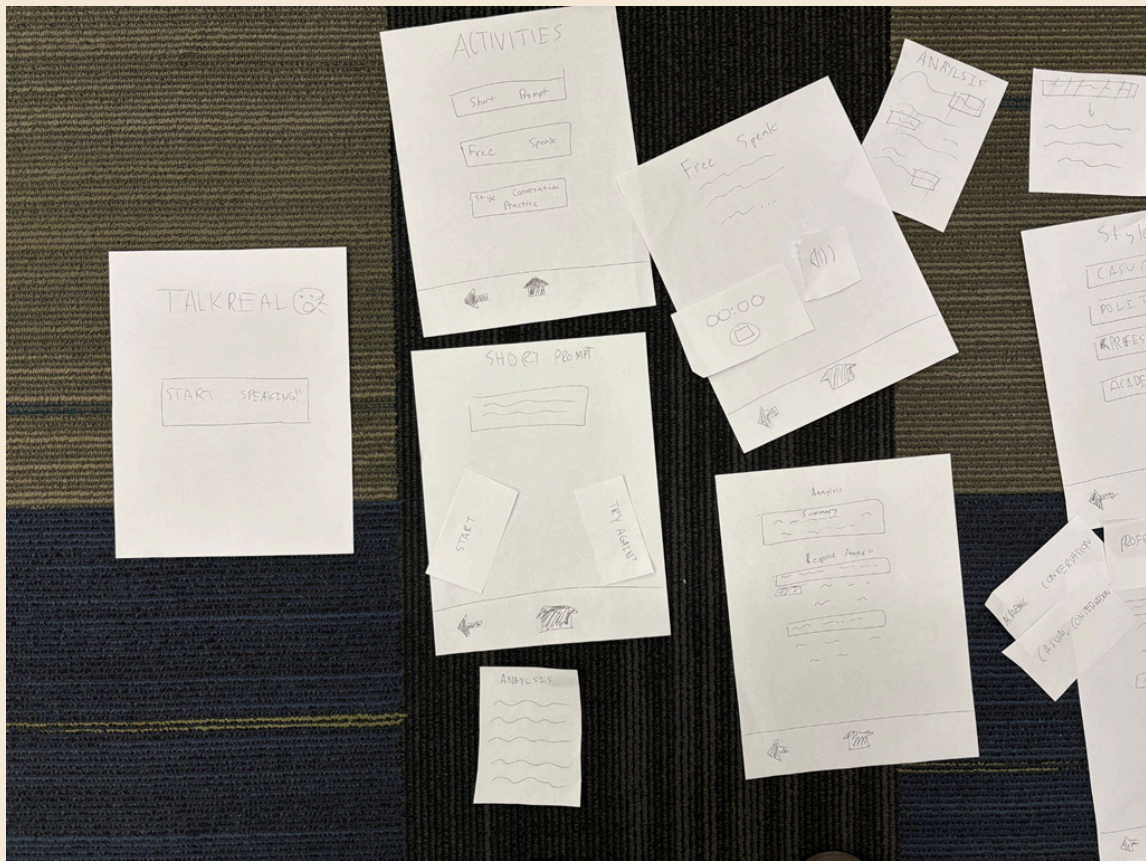


Figure 19. One low-fi paper prototype setup shows how the team physically arranged screens and movable pieces to simulate the core app flow.

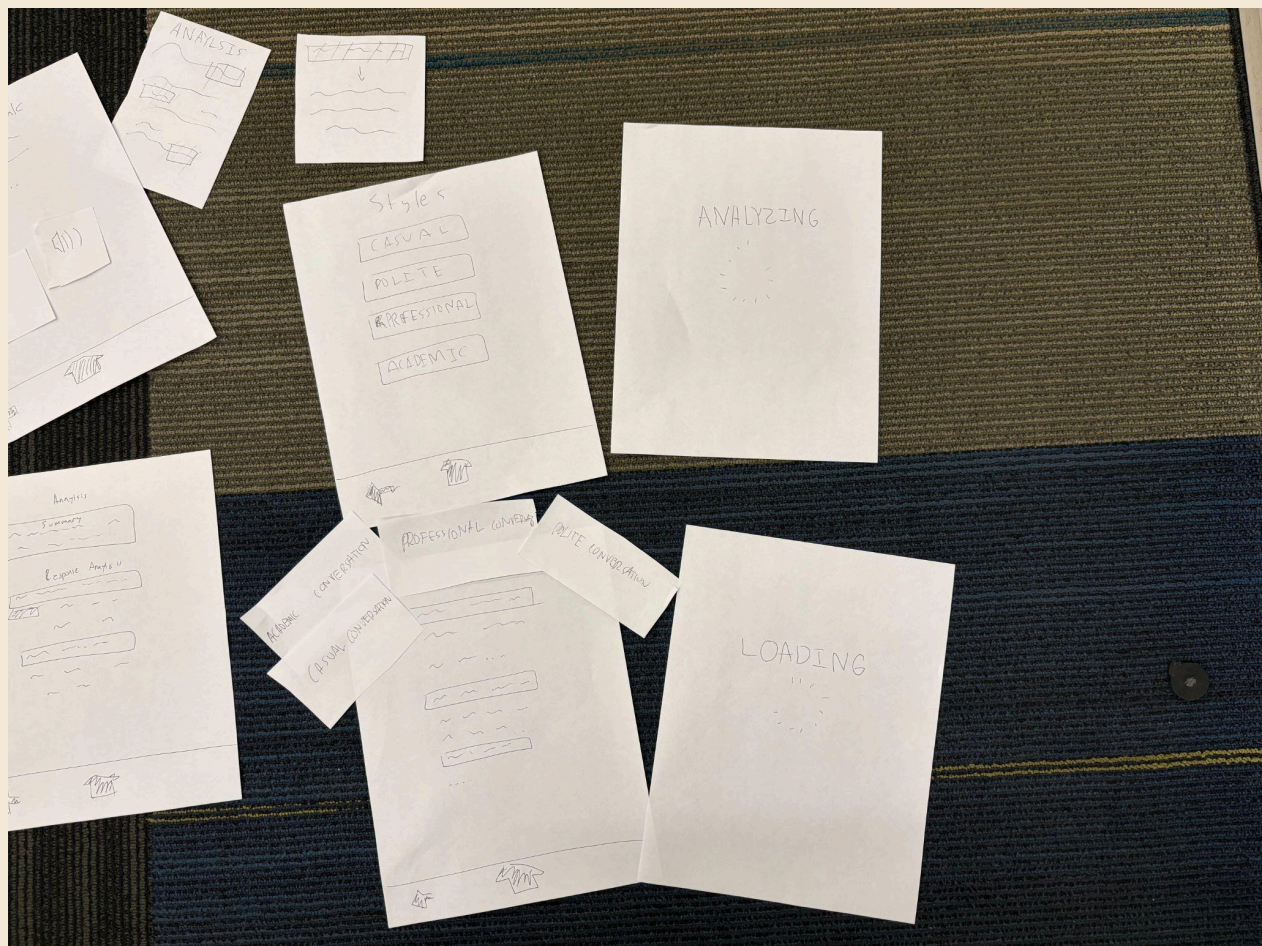


Figure 20. A second paper-prototype arrangement captures additional conversation and analysis states used during testing.

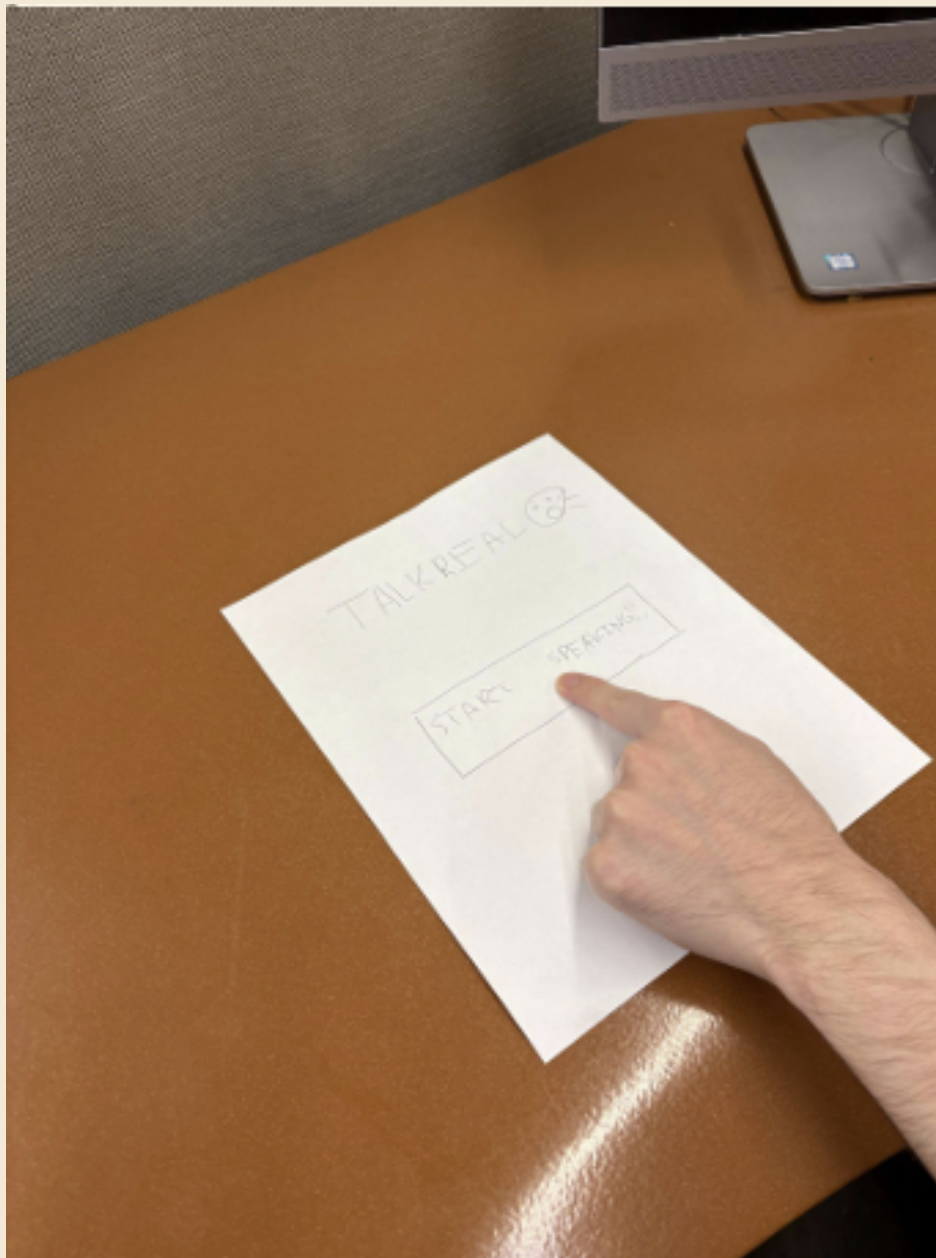


Figure 21. A participant interacting with the paper prototype during usability testing. This setup made it possible to observe pointing, hesitation, and facilitator intervention in real time.

	Assists needed from facilitator	Misclicks	Ease-of-use (1-5)
Frieren	2	3	3
Mahito	1	0	4

Figure 22. A compact testing summary shows the two participants' facilitator assists, misclicks, and ease-of-use ratings.

From Medium-Fi to High-Fi

The medium-fi prototype was strong enough to support peer heuristic evaluation, which then drove the most substantial improvements in the final design. Across the heuristic feedback we received, the most serious violations clustered around user control, error prevention, consistency, and unfinished system states. The high-fi redesign focused on making TalkReal feel calm, trustworthy, and production-like while removing forced automation and dead ends.

Below are the major severity 3 and 4 issues raised by evaluators, along with our design response:

- **Immediate transition from recording to analysis across all tasks (S3).** We added a review-before-analysis screen so users can replay, inspect, edit, or re-record before committing to feedback.
- **Short Prompt ending with no warning (S3).** We added a visible countdown and a ten-second warning state so users can wrap up naturally.
- **Free Response lacking pause and resume (S3).** We added pause and resume controls to support pacing and reduce pressure.
- **Progress screen as a placeholder (S4).** We replaced the placeholder with a believable progress dashboard that shows streaks, trends, recent scores, and practice distribution.
- **No onboarding for first-time users (S4).** We added a short onboarding carousel that explains what TalkReal does, how the three activities differ, and what kind of feedback users should expect.
- **No transcript editing before analysis (S3).** We added transcript correction during the review step to reduce the damage caused by speech recognition errors.
- **Users could not exit results immediately (S3).** We redesigned the results layout so primary next actions are available near the top instead of being blocked behind deeper analysis sections.
- **The Short Prompt reload icon was unclear (S3).** We relabeled and contextualized the action so users understand it as requesting a new prompt.
- **Opening one feedback dropdown opened all of them (S4).** We redesigned the results screen so each accordion opens independently and no longer yanks the user to the wrong part of the page.
- **Open feedback items could not be closed (S3).** We made open and close states explicit and reversible.
- **Free Response results had overlapping text (S3).** We rebuilt card spacing and dynamic sizing so explanations expand cleanly.
- **The conversation end button scrolled off-screen during long conversations (S4).** We fixed the top bar in place so the user always has access to navigation and a clear exit path.

This med-fi to high-fi shift also changed the tone of the product. In medium-fi, the app worked as a concept demonstration. In high-fi, it began to feel like a real learning tool. The interface became more consistent, more transparent, and more respectful of user control. That was especially important because our users were already anxious about speaking. Any extra friction from the interface amplified the very problem we were trying to solve.

6c.i. Additional High-Fi Improvements

Alongside those major fixes, we made several supporting revisions that improved cohesion:

- Renamed **Stylish Talking** to **Conversation Practice** so the feature title better matches the task.
- Implemented a meaningful **Profile** page instead of leaving it empty.
- Redesigned the **Home** screen to surface practice history and motivation at a glance.
- Added a **How Scoring Works** explanation so the feedback feels more trustworthy.
- Removed redundant buttons that sent users to the same destination under different labels.
- Added confirmation dialogs at high-stakes exit points to reduce accidental loss of progress.

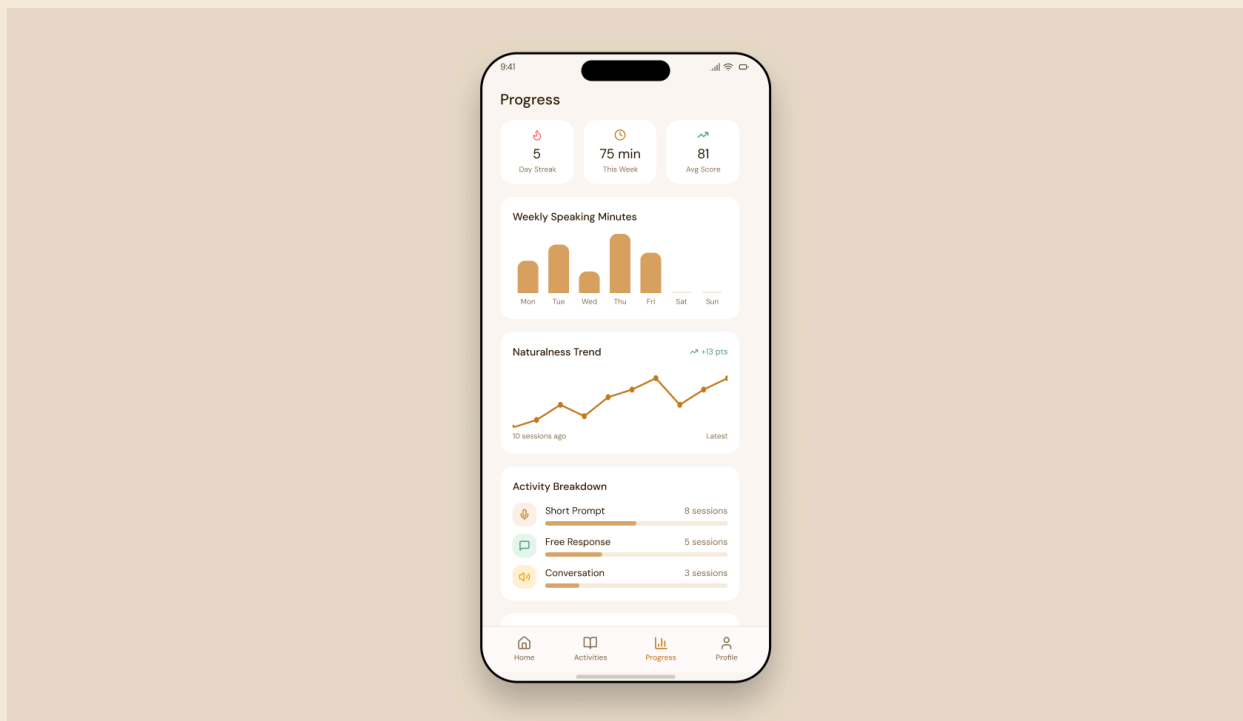


Figure 23. High-fi progress dashboard replacing the earlier placeholder state.

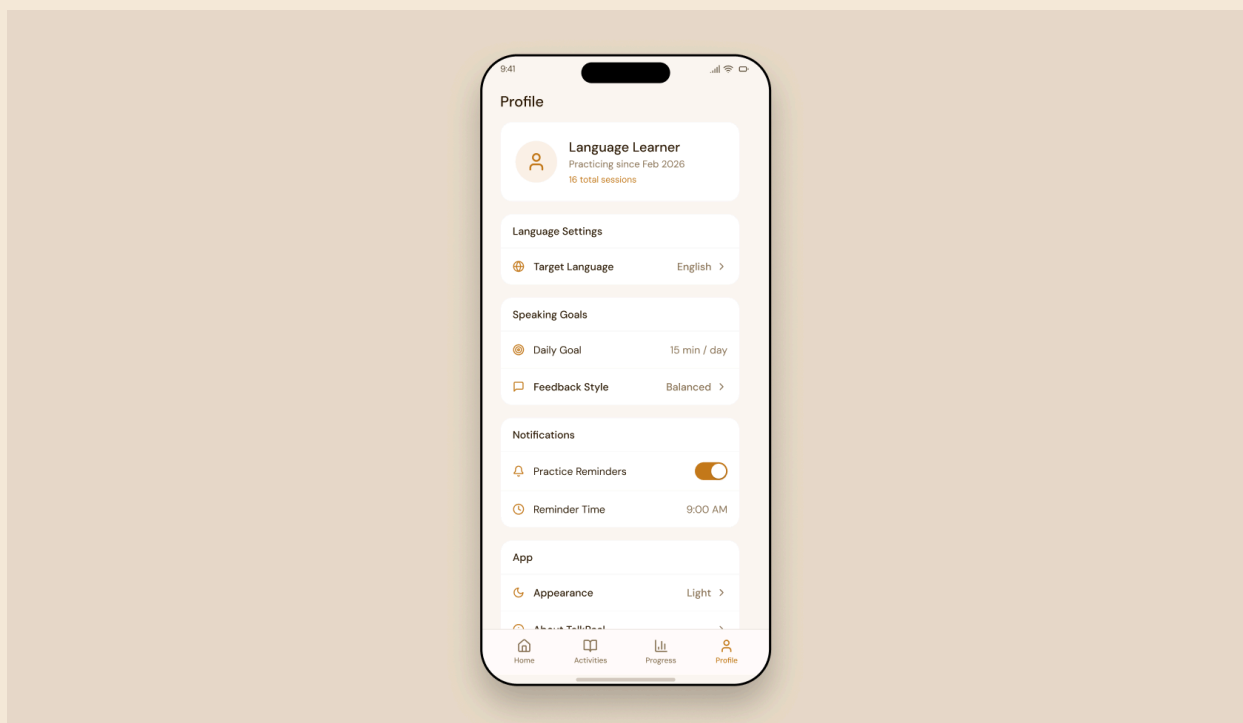


Figure 24. High-fi profile screen expanding the product beyond the core task flows.

7. Final Prototype

7a. Hard-Coded Data Used

Even in the final prototype, much of the data remained scripted so we could test the interaction model without a live backend. This included fixed prompts, scripted transcripts, pre-written analysis cards, static naturalness scores, simulated loading states, predetermined conversation turns, and dashboard data such as streaks or weekly minutes. In earlier prototype documentation, for example, the Short Prompt scenario remained a cafe-ordering prompt, Free Response used a fixed monologue, and the style-based conversation used prewritten exchanges for each tone.

Using hard-coded content was appropriate for this phase because our main research question was not model accuracy. It was whether users understood the flow, trusted the feedback structure, and felt motivated to continue practicing. By stabilizing the content, we could focus testing on clarity, learnability, and perceived effort.

7b. AI Tools Used and Why

TalkReal is inherently an AI-centered product concept, but our prototype intentionally simulated AI behavior rather than connecting to a live speech model or conversation backend. We used hard-coded outputs to stand in for speech recognition, naturalness analysis, and AI replies so that we could evaluate interaction design before introducing the variability of real model output.

This choice was justified for two reasons. First, a stable scripted prototype let us compare flows and feedback structures consistently across participants. Second, it prevented model errors, latency swings, or recognition failures from masking design issues that we still needed to solve first, such as user control, discoverability, and trust. In other words, we treated AI capability as the product premise, but we treated the quarter's prototypes as interaction-design artifacts rather than technical ML deployments.

8. Reflection and Next Steps

8a. Main Learnings

The biggest lesson from this quarter was that confidence is not a side effect of language learning. For our users, confidence is the design problem. Grammar and vocabulary matter, but what actually blocked speaking was fear of judgment, fear of sounding unnatural, and uncertainty about whether AI feedback would transfer to real life.

We also learned that fidelity changes what kind of feedback becomes possible. Early needfinding helped us choose the right problem. Experience prototypes helped us test emotional and behavioral assumptions quickly. Low-fi testing then exposed navigation and comprehension issues that would have been easy to overlook if we had jumped straight into polish. Finally, heuristic evaluation showed that a concept can be promising and still feel unsafe if the interface removes control or hides important actions.

Another major takeaway was that supportiveness needs to be designed very intentionally. Small choices such as adding a review step, allowing transcript edits, showing a ten-second warning, or surfacing a clear exit button all communicate respect for the learner. In a speaking app for anxious users, those details are not cosmetic. They are part of the core value proposition.

8b. Future Work

If we had more time, the next step would be to move from scripted prototype behavior toward a functional end-to-end system. That would include real speech recognition, live naturalness analysis, and deeper testing of how users respond to imperfect or uncertain AI feedback. We would also want to test multilingual transfer more directly, especially for languages where politeness, register, and context heavily shape naturalness.

Beyond technical implementation, we would expand the learning loop. Possible next steps include adaptive difficulty, richer progress tracking, saved history with replayable transcripts, personalized feedback styles, and scenario packs for common real-life tasks such as job interviews, doctor appointments, small talk, and travel. We would also revisit the balance between speaking freedom and structure, since different learners may want different degrees of prompting, pacing, and realism over time.